

Qi Ding

Ph.D. Candidate in Management Science and Engineering
Institute of Energy, Environment, and Economy, Tsinghua University
dingq23@mails.tsinghua.edu.cn · ding-qi.top · github.com/dingqi01

Education

Tsinghua University

2023–present

Ph.D. in Management Science and Engineering, Institute of Energy, Environment, and Economy

Tongji University

2018–2023

B.Eng. in Vehicle Engineering, School of Automotive Studies

Research Interests

Transport decarbonization and integrated energy systems; electric vehicles, charging, and power-system interactions; hydrogen and low-carbon fuels; optimization and data-driven analysis for energy and climate policy.

Publications

- **Ding, Q.**, Ren, J., Zhang, S., & Chen, W. (2026). The role of shared autonomous electric vehicles in decarbonizing China's passenger transport sector. *Applied Energy*, 422, 128259. [DOI]
- **Ding, Q.**, & Chen, W. (2026). Retired Battery Supply and Energy Storage Demand in China: A Provincial-Scale Assessment. *Energy Proceedings*, 65. [DOI]
- Chen, W., Zhang, S., Zhang, Q., Ren, J., & **Ding, Q.** (2025). Assessing China's Province-Coordinated Power System Carbon-Neutral Transition Pathway. *Journal of Energy and Climate Change*, 1(1), 1–15. [DOI]

Manuscript Under Review

- **Ding, Q.**, et al. Aligning Retired EV-Battery Supply with Grid-Storage Demand through Interprovincial Coordination in China. Manuscript under review at *Applied Energy*.

Selected Research Projects

- **National Key R&D Program of China — CO₂-EOR and Long-term Sequestration** Nov 2024–present
Core member. *Techno-economic assessment of CCUS pathways, sequestration benefits, and cost curves.*
- **European Commission, DG CLIMA — COMMITTED Project** Sep 2024–present
Research contributor. *Multi-model comparison of climate-policy portfolios and transition pathways.*
- **Energy Foundation China — China Energy System Model Comparison** Sep 2023–present
Research member. *Comparison of model assumptions and decarbonization pathways across energy-system models.*
- **The World Bank Group — Green China** Sep 2023–Jun 2024
Researcher. *China TIMES 2.0 analysis of low-carbon power and industry transition pathways.*
- **CNPC Economics & Technology Research Institute — Transportation Energy Refueling Facility Upgrade** Jul–Dec 2023
Research member. *Assessment of multi-energy hubs spanning charging, battery swapping, and hydrogen.*
- **China Automotive Technology and Research Center — Commercial-Vehicle Carbon-Neutrality Roadmap 1.0** Oct 2022–Jul 2023
Core member. *Assessed electrification, hydrogen, and biofuels for the clean-energy-supply workstream.*

Methods and Skills

Integrated assessment, optimization, techno-economic analysis, GIS, and data-driven modeling; Python, GAMS, MATLAB, TIMES, and PyPSA; English (CET-6: 641) and German (Band 6: Excellent).

Selected Honors

National Scholarship (2021, 2022); Shanghai Outstanding Graduate (2023).